

Bubble Plume Integral Model for Line-Source Diffusers in Ambient Stratification

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Abstract: We developed an integral plume model to simulate the behavior of bubble plumes generated from line-source geometry in stratified ambient reservoirs by adapting the double-plume integral model developed for point-sources to a line plume. The model, based on top-hat velocity and buoyancy profiles, uses an Eulerian integral modeling approach and predicts the hydrodynamic, chemical, and thermodynamic behavior of the bubbles using a discrete bubble model. Existing integral models for line-source bubble plumes consider only the upward motion of bubbles and entrained water. To accurately predict intrusion formation, mixing patterns, and efficiency of bubble plumes in stratified environments, the downward flow of plume fluid from the maximum extent of plume rise to the trap height should also be included. To solve this problem, we presented a derivation of a continuously peeling double-plume model for line plumes and calibrated the model peeling factor to data for trap height in two stratified reservoirs. We applied the calibrated model to predict the gas transfer and vertical fluxes of oxygen from an oxygenation system in Carvins Cove reservoir and compared the predictions of the double-plume model to that using a standard single-plume model. We showed that the amount and the vertical distribution of entrainment into the plume differs in the double-plume model compared to the single-plume; hence, the type of model (double-plume or single-plume) would affect the results of simulations in coupled reservoir circulation models. **DOI: 10.1061/(ASCE)HY.1943-7900.0001885.** © 2021 American Society of Civil Engineers.

Introduction

Engineered bubble plumes with a line-source geometry are used widely in natural systems, yet the development of integral models to predict their behavior has lagged behind the development of models for point sources. In most natural reservoirs, the density of the ambient water is stratified, and this stratification acts to resist the vertical upwelling generated by the bubble plume, eventually resulting in detrainment of water from the bubble plume and formation of lateral intrusions. For point-source plumes, a double-plume integral modeling approach has been developed (McDougall 1978; Asaeda and Imberger 1993; Socolofsky et al. 2008), consisting of an upward-rising inner plume of bubbles and entrained water surrounded by a downward-flowing outer plume of water detrained from the inner plume. In line-source bubble plumes, a double-plume integral model formulation is lacking, and existing models only account for the upward pumping of water by the bubbles; hence, they cannot accurately predict intrusion dynamics, an important process affecting the mixing patterns in a reservoir. Here, we

derive a double-plume integral model of a line bubble plume and calibrate the model's detrainment parameter to data from Singleton et al. (2007) in stratified Spring Hollow reservoir in Virginia. We further use the calibrated model to study oxygen dynamics in Carvins Cove reservoir in Virginia during targeted field experiments in 2011 and 2013 (Rezvani 2016; Bierlein et al. 2017). The new double-plume integral model for line-source bubble plumes is important to accurately predict entrainment, upwelling, and intrusion dynamics from line bubble plumes in stratified ambient reservoirs.

Examples of engineered uses of line-source bubble plumes include plumes for aeration, oxygenation, and destratification in lakes and reservoirs; for prevention of saline water intrusions; for pneumatic breakwaters; to arrest the surface spreading of oil; and to contain sound pollution from subsea explosions and construction activities, among many others. In most of these cases, the upward flow of water entrained with the bubbles is critical to the success of the system because it either generates the desired mixing (destratification and aeration) or it creates a barrier to crossflow (pneumatic breakwaters and oil herding). In bubble curtains for sound suppression, the upwelling velocity in the plume is also important because it sets the residence time of the bubbles in the water column. When ambient density stratification interrupts this vertical pumping of entrained water, models need a mechanism to account for intrusion formation.

For bubble plumes from point-source geometry, intrusion dynamics are handled in integral models using some variation of the double-plume model, introduced by McDougall (1978) and simplified by Asaeda and Imberger (1993). These original double-plume models accounted for detrainment from the inner plume and formation of the outer plume at discrete points along the vertical trajectory of the plume. Crounse et al. (2007) introduced a continuous detrainment algorithm. This continuous peeling model was also adapted by Socolofsky et al. (2008) and validated to a wide range of data in the literature. Yang et al. (2016) further tested the continuous peeling approach using results from large eddy simulation (LES) of a bubble plume in linear density stratification and

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proposed a modified formulation of the detrainment algorithm that accurately accounted for the force balance in the inner plume. Hence, double-plume integral models for point-source bubble plumes are well established in the literature.

However, double-plume models have not always been used for integral models of bubble plumes in reservoirs. Early work adapted single-phase integral plume models to the multiphase nature of bubble plumes (e.g., Kobus 1968; Cederwall and Ditmars 1970; Fanneløp and Sjoen 1980; Milgram 1983; Wüest et al. 1992). Models for lake and reservoir aeration using line-source bubble plumes developed out of these earlier models. Hussain and Narang (1984) presented a model for a line-source bubble plume in stratification and allowed the bubbles to exit the top of the plume, stopping the model integration at the point of maximum plume rise, where the vertical momentum in the entrained water in the plume declines to zero. McGinnis et al. (2001) expanded this model with bubble dissolution dynamics following an approach similar to Wüest et al. (1992). Singleton et al. (2007) further applied the McGinnis et al. (2001) model for a linear bubble plume and validated the model to data in Spring Hollow reservoir for the depth of maximum plume rise, the temperature of water in the plume at the top of the simulation, and dissolved oxygen profiles through the plume.

The model of Singleton et al. (2007) could predict the extent of the plume well, but temperature predictions deviated from the measured data near the top of the plume due to the missing detrainment and intrusion dynamics in the model. Indeed, the inner-plume and outer-plume dynamics involved in intrusion formation have been observed in many reservoir studies (e.g., Schladow 1992; Lemckert and Imberger 1993; McGinnis et al. 2004). As the outer plume descends from the point of maximum plume rise to the trap depth, it shrouds the inner plume, effectively cutting it off from the ambient reservoir above the trap depth. Though single-plume models for line-source bubble plumes can be used to design the gas flux and bubble sizes for oxygenation systems, they could provide better estimates of induced circulation if coupled to a double-plume integral model approach to capture detrainment and intrusion formation.

To bridge this gap, we adapted double-plume integral models from point-source bubble plumes to those with line-source geometry. We began with an inner plume model based on McGinnis et al. (2001) and Singleton et al. (2007) with updated equations-of-state and dissolution models for the bubbles from Dissanayake et al. (2018). To initialize the outer plume, we modified the detrainment algorithm of Crounse et al. (2007) to the two-dimensional, line-source geometry and present an entrainment model for interaction between the inner and outer plume also adapted to the two-dimensional case. To calibrate the detrainment parameter, we used data from Singleton et al. (2007) for Spring Hollow reservoir. Finally, we validated the model using data collected from field campaigns in Carvins Cove Reservoir in 2011 and 2013 (Rezvani 2016; Bierlein et al. 2017).

This paper is organized as follows. The adapted model equations are presented in the “Model Development” section, followed by a dimensional analysis to define the characteristic scales for length, time and velocity. The model calibration and validation presents the model application in Spring Hollow and Carvins Cove Reservoirs. We discuss the differences between the single-plume and double-plume model results for linear bubble plumes in the section relating to the effect of the double plume model on plume behavior. Finally, we report the summary and conclusions.

Model Development

We based our new, double-plume integral model for a line-source, or linear, bubble plume on the governing equations for a

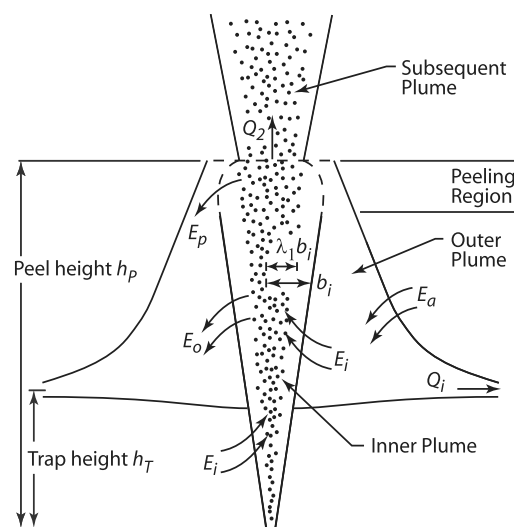


Fig. 1. Schematic diagram of the double-plume model. (Reprinted with permission from Socolofsky et al. 2008, © ASCE.)

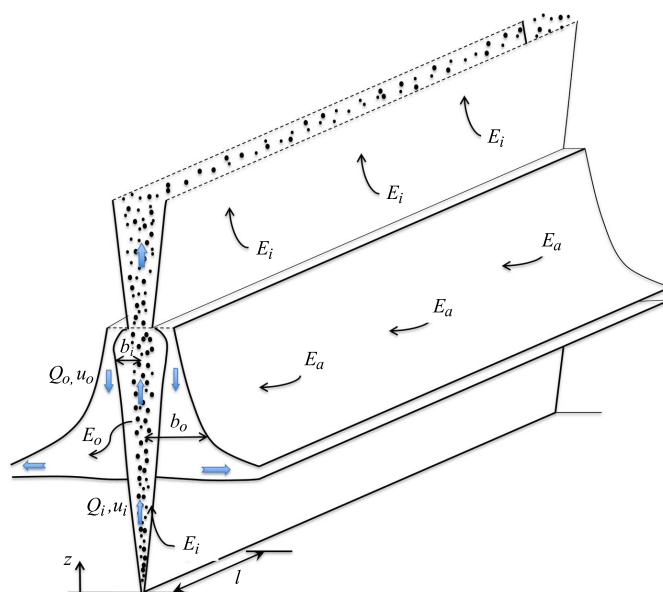


Fig. 2. Schematic diagram of the double-plume model for a line-plume in three-dimensional view.

point-source, or circular, bubble plume in Asaeda and Imberger (1993) using the continuous peeling approach of Crounse et al. (2007). When bubbles are released underwater, they rise while entraining the surrounding water. In a density-stratified reservoir, the bubble plume carries heavy water entrained at lower levels in the plume to the upper water column, where the ambient fluid is lighter. As the upward flow of water and bubbles is slowed by the stratification, this heavy, entrained water starts to detrain (peel) from the bubble plume, creating an outer plume of water moving in the opposite direction to the rising bubbles (Figs. 1 and 2). The top of the detrainment region is known as the peeling height (Socolofsky and Adams 2005; Socolofsky et al. 2008). Residual bubbles may continue to rise in the water column, and the entrainment and detrainment process starts over again. In the double-plume integral model, the rising plume of bubbles and entrained water is called the inner

plume, and the detrainment plume of water is called the outer plume. Fig. 1 presents a schematic of the double-plume model, including several definitions and scales.

Fig. 2 shows a three-dimensional sketch of the double-plume integral model applied to a linear source geometry. For an idealized linear plume, we assumed the source extends infinitely along the source axis (y-direction); hence, the state-space fluxes are presented in terms of per unit length l along the plume. Using top-hat velocity profiles, the integral inner and outer plume fluid flow rates Q_i and Q_o can be defined as

$$Q_i = \int_0^l \int_{-\infty}^{+\infty} u_i dx dy = 2b_i l u_i \quad (1)$$

$$Q_o = \int_0^l \int_{-\infty}^{+\infty} u_o dx dy = 2(b_o - b_i) u_o \quad (2)$$

where x is in the horizontal plane normal to the source; z is positive upward; u_i and u_o = inner and outer top-hat plume water velocities; b_i and b_o = inner and outer plume half-widths; and l is taken as a unit width. Here, we have made the dilute-plume assumption in which the void fraction of the bubbles can be neglected when computing the inner plume volume flux (Socolofsky et al. 2008). Similarly, the integral momentum flux of the inner (M_i) and outer (M_o) plumes can be derived as

$$M_i = \int_0^l \int_{-\infty}^{+\infty} u_i^2 dx dy = 2b_i l u_i^2 \quad (3)$$

$$M_o = \int_0^l \int_{-\infty}^{+\infty} u_o^2 dx dy = 2(b_o - b_i) l u_o^2 \quad (4)$$

With these definitions of the state variables, the equations for conservation of mass, momentum, salinity, heat, and dissolved species flux (Dissanayake et al. 2016) for the inner and outer plumes in an Eulerian reference frame closely follow the equations presented for a point-source release in Socolofsky et al. (2008) with a few revised parameter definitions to account for the two-dimensional nature of the line-source plume, described as follows.

In a double-plume integral model based on the entrainment hypothesis, four different entrainment fluxes are typically considered (Figs. 1 and 2). These are from the ambient water to the inner plume E_i , from the ambient water to the outer plume E_a , from the outer plume to the inner plume (also E_i), and from the inner plume to the outer plume E_o . These entrainment fluxes for a line plume model are defined as follows:

$$E_i = \alpha_i 2l(u_i + c_1 u_o) \quad (5)$$

$$E_o = -\alpha_o 2l u_o \quad (6)$$

$$E_a = -\alpha_a 2l u_o \quad (7)$$

where α_i , α_o , and α_a = entrainment coefficients; and c_1 = parameter with valid values of -1 or 0 to select between two different inner-plume entrainment models (see Asaeda and Imberger 1993; Socolofsky et al. 2008). The equation for conservation of momentum flux for the inner plume includes the local kinematic buoyancy flux of the entrained water B_i and the contained bubbles B_b . For a line plume using the dilute-plume assumption, B_i is

$$B_i = g \left(\frac{\rho_a - \rho_i}{\rho_r} \right) l b_i u_i \quad (8)$$

where g = acceleration of gravity; ρ_a = density of the ambient water at a given depth; ρ_i = local density of the inner plume water;

and ρ_r = reference density, which is taken as the ambient density at the middepth of the water body. Likewise, B_b is given by

$$B_b = g \frac{(\rho_a - \rho_b)}{\rho_r} \chi l b_i u_i \quad (9)$$

where χ = local void fraction, expressed for a line-source plume as

$$\chi = \frac{2\dot{m}_b}{\rho_b \sqrt{\pi} \lambda_1 l b_i (u_s + u_i / \sqrt{1 + \lambda_1})} \quad (10)$$

where $\dot{m}_b$ = mass flux of the bubbles; λ_1 = ratio of the bubble column width to the velocity width; and u_s and ρ_b = slip velocity (terminal rise velocity) and the density of the bubbles, respectively.

The integral model formulation in Socolofsky et al. (2008) tracks the evolution of salinity, temperature, and dissolved species and computes the density of the local water and injected gas using separate equations of state. Water properties were computed from the seawater equation of state in Gill (1982). Bubble properties were computed using the discrete-bubble model approach (Wüest et al. 1992) in which bubbles of a characteristic size are tracked along the plume centerline. We used the Peng-Robinson equation of state for real gas bubbles as described in Dissanayake et al. (2018). To solve the corresponding set of governing equations, we applied an ordinary differential equation (ODE) solver for systems of stiff equations as in Socolofsky et al. (2008) and Dissanayake et al. (2018).

Continuous Peeling Model

To close the set of model equations coupling the inner to the outer plume, we used a continuous-peeling approach to detrainment. Here, we chose the original continuous-detrainment algorithm, introduced by Crounse et al. (2007) and validated for point-source plumes in Socolofsky et al. (2008), in which the detrainment flux E_p is given as

$$E_p = -\varepsilon_p \left(\frac{u_s}{u_i} \right)^2 \left(\frac{B_i}{u_i^2} \right) \quad (11)$$

where ε_p = calibration parameter. Peeling becomes more intense as the inner plume decelerates and u_i approaches zero; smaller values of ε_p delay peeling, resulting in more abrupt outer plume formation.

Initial Conditions for the Model

Initial conditions for the integral model require an estimate of the vertical velocity of entrained water at the end of the zone-of-flow-establishment (ZFE). Several approaches for point-source plumes exist in the literature (Socolofsky et al. 2008), and here, we adopted a formulation dependent on the densimetric Froude number Fr_0 as proposed by Wüest et al. (1992) for point-source bubble plumes and adapted by Singleton et al. (2007) to line sources. This method has since been successfully used by Socolofsky et al. (2008), McGinnis et al. (2004), and Dissanayake et al. (2018) for point sources. The line-source version of this initial condition derived by Singleton et al. (2007) is

$$Fr_0 = \frac{u_i}{\sqrt{\lambda_1 2b_i g(\rho_a - \rho_m) / \rho_m}} \quad (12)$$

where ρ_m = density of the mixture of bubbles and water in the inner plume [$\rho_m = \chi \rho_b + (1 - \chi) \rho_i$]; and b_i is taken as the half-width of the diffuser. This approach assumes that Fr_0 approaches a constant near the source. Eq. (12) is iteratively solved

for u_i (which affects the value of χ , hence, ρ_m) until the selected constant value of Fr_0 is achieved. We adopted a value of $Fr_0 = 2$, following suggestions by Fanneløp and Webber (2003) and Singleton et al. (2007) to prevent an increase in velocity and contraction of the plume close to the source. To initialize outer plume segments, we likewise used a Froude-number approach, as described in Socolofsky et al. (2008) and used the value $Fr_{0,o} = 0.2$.

Entrainment Coefficient in the Model

Entrainment coefficients used in the model followed recommendations from the literature on single-phase and multiphase line plumes. In the following, we discuss in brief the historical context of entrainment coefficients followed by the reported values of entrainment coefficients in the literature for models assuming a Gaussian velocity profile. Top-hat values of entrainment coefficients were obtained by multiplying Gaussian values by $\sqrt{2}$ (Fanneløp et al. 1991; Singleton et al. 2007; Wüest et al. 1992).

Entrainment coefficients used in models are developed from both a theoretical and experimental basis. In integral jet and plume models, entrainment is widely used in the closure formulation, stemming from early studies by Priestley and Ball (1955) and Morton et al. (1956) with the assumptions of fully developed plumes. Governing equations used by Morton et al. (1956) considered the volume-flux, the momentum flux, and the buoyancy flux in their formulation and suggested an entrainment coefficient based on volume conservation, whereas Priestley and Ball (1955) used flux of mean kinetic energy in their solution instead of volume flux and relied on parameterization based on the production of turbulent kinetic energy for closure (van Reeuwijk and Craske 2015). Priestley and Ball (1955) found that the entrainment coefficient is proportional to the local Richardson number. Even though there has been criticism of the Priestley and Ball (1955) formulation for neglecting the continuity equation, Fox (1970) compared both models in his analysis by simultaneously considering volume-flux, momentum flux, buoyancy flux, and the flux of mean kinetic energy. They showed that entrainment is a function of the mean kinetic energy, the form of the similarity profiles, and the local mean densimetric Froude number, for both axisymmetric and plane plumes in a stratified ambient. Kaminski et al. (2005) revisited the work of Priestley and Ball (1955) to derive the entrainment behavior for the initial flow regime of a volcanic plume in which buoyancy acts against the momentum. Their new theoretical parameterization of entrainment is a function of the local (negative) Richardson number. In their formulation, van Reeuwijk and Craske (2015) considered a restriction on the entrainment coefficient by the equation for mean kinetic energy and referred to this as the entrainment relation. Their entrainment relation suggests that the entrainment coefficient includes contributions from turbulence production, buoyancy, and changes in the profile coefficients associated with the production of turbulent kinetic energy. Profile constants in these formulations need to be defined for different flows and estimated from laboratory experiments. Based on their analysis of data for pure jets and plumes in unstratified environments, van Reeuwijk and Craske (2015) indicate that the larger entrainment in plumes is mainly associated with the behavior of the mean flow and not with buoyancy-enhanced turbulence. The latter is usually considered the reason for the larger entrainment in plumes than in jets (Kaminski et al. 2005).

Entrainment coefficients for jets and plumes presented in the literature have a range of values. For single-phase plumes, Fanneløp and Webber (2003) applied an integral model and found an entrainment coefficient of 0.08 by fitting the Ricou and Spalding (1961) entrainment model with data from Liedtke and Schatzmann (1997). Based on experimental data, Lee and Emmons (1961) determined

the line-plume entrainment coefficient to be 0.16 and stated its agreement with a similar study by Rouse et al. (1952). In an experimental study by Yuan and Cox (1995), the entrainment coefficient was expressed as a function of centerline temperature and velocity, and they reported that constant values in the literature varied from 0.12 to 0.16. Lee and Chu (2003) concluded for line sources that the pure jet entrainment coefficient was 0.053 and a pure plume was 0.103; Jirka (2006) similarly reported values of 0.0625 for pure jets and 0.125 for pure plumes. Hence, some of the variability in the literature may depend on the setup used, the degree to which momentum played a role in an experiment, and best-fit entrainment coefficients in single-phase, Gaussian line-plume models range from 0.05 to 0.16.

A few studies also considered air bubble plumes in line-source geometry. Based on experiments in Kobus (1968), and Cederwall and Ditmars (1970) used an integral model to show that the entrainment coefficient may vary between 0.09 and 0.12, and they proposed the possibility that there is an asymptotic value of about 0.16. Hussain and Narang (1984) suggested an entrainment coefficient of 0.09 for two-dimensional, two-phase plumes. Singleton et al. (2007) used a value of 0.11 following Fanneløp et al. (1991) and Fanneløp and Sjoen (1980). As with point-source bubble plumes, the entrainment coefficient for a line-source bubble plume is not expected to be strictly constant (e.g., Milgram 1983), and the studies here reported a similar range of values to those in the single-phase literature for line-source plumes.

Because of the uncertainty in the entrainment coefficient for multiphase plumes and the similarity of its value to single-phase plumes, we followed Dissanayake et al. (2018) and applied single-phase entrainment coefficients to the bubble plumes. Following Jirka (2004, 2006) and Lee and Chu (2003), we computed the effective entrainment coefficient of a buoyant jet using a Froude number approach that appropriately combined the pure-jet and pure-plume entrainment coefficients, depending on the local conditions of the inner plume (see Dissanayake et al. 2018 for a detailed application to point-source bubble plumes). Harleman et al. (1973) and Jirka (2006) provided this functional relationship for entrainment coefficient in terms of Froude number to accommodate the transition between local negative and positive buoyancy effects on plumes in stratified environments, following the energy conservation equation in Fox (1970). We adopted the jet and plume entrainment coefficient values for line-source plumes from Jirka (2006), adjusted to a top-hat velocity profile, with an asymptotic, pure-plume densimetric Froude number of $Fr_{lp} = 3.6$.

The entrainment coefficients α_o and α_a in Eqs. (6) and (7) were assumed to be 0.11 for the present top-hat model (Socolofsky et al. 2008). None of the values of these entrainment coefficients were varied during calibration or validation of the model. Instead, the only available model fit parameter was the coefficient of the continuous peeling flux ε_p in Eq. (11).

Dimensional Analysis

To present experimental and model results in nondimensional form, we adapted the dimensional analysis of Socolofsky and Adams (2005) for a point-source plume to a line-source bubble plume. First, we considered the characteristic scales for velocity, length, and time. In contrast to a point-source plume, in which the velocity scales as $(B_r/z)^{1/3}$, where B_r is buoyancy flux of bubbles in a point source plume, the velocity in a line-source plume is constant and scales as $B^{1/3}$. A characteristic length scale of a line-source plume scales as $(B/N^3)^{1/3}$, where B is the buoyancy flux in a line source plume, $N = \sqrt{(g/\rho_r)(\partial\rho_a/\partial z)}$ is the Brunt-Vaisala

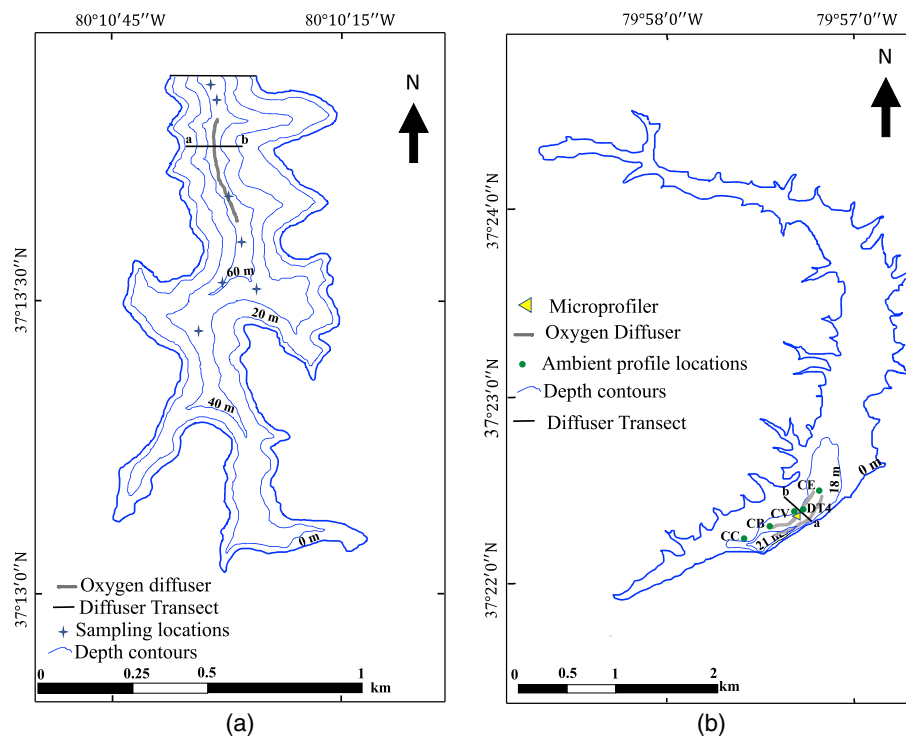


Fig. 3. Maps of (a) Spring Hollow Reservoir (data from Singleton et al. 2007); and (b) Carvins Cove Reservoir and the locations of the diffusers (data from Rezvani 2016; Bierlein et al. 2017).

buoyancy frequency, and a characteristic time is N^{-1} . These parameters are essentially the same for single-phase and multiphase line plumes.

To introduce the bubbles into the dimensional analysis, most papers in the literature use the slip velocity u_s (Socolofsky and Adams 2005; Bombardelli et al. 2007). Dimensional analysis of the four parameters B , N , u_s , and z with two dimensions of length and time yields two independent nondimensional variables; these are

$$U_N = u_s/B^{1/3} \quad (13)$$

$$Z_N = z/(B/N^3)^{1/3} \quad (14)$$

Socolofsky and Adams (2005) show that many plume properties are functions of the nondimensional slip velocity U_N , and generally, we can replace z in the dimensional analysis with any other state variable ϕ and seek correlations of the form

$$\phi/\theta(B, N) = f(U_N) \quad (15)$$

where $\theta(B, N)$ = proper combination of B and N to nondimensionalize ϕ . We used this approach in the analysis herein.

Model Calibration and Validation

Because the formulation of the inner-plume model and selected entrainment coefficients follow similar models for linear bubble plumes in the literature, we focused the model calibration and validation on the detrainment model, which includes the peeling parameter ε_p and the entrainment algorithm between the inner and outer plumes, selected by the parameter c_1 in Eq. (5). Socolofsky et al. (2008) presented a careful validation of the

double-plume entrainment algorithms and recommended a value of $c_1 = 0$, which we also used here.

Because of the lack of laboratory data for line-source bubble plumes in a stratified ambient fluid, we used field data to calibrate and validate the detrainment models and to determine the appropriate value of ε_p . Singleton et al. (2007) presented measurements of temperature and oxygen concentration versus depth in Spring Hollow reservoir, a drinking water reservoir in Virginia with hypolimnetic oxygen concentration managed by a linear bubble plume [Fig. 3(a)]. We also made similar field observations in nearby Carvins Cove reservoir, also oxygenated by a line-source bubble plume diffuser [Rezvani 2016; Bierlein et al. 2017; Fig. 3(b)]. In both studies, the airflow rate of the diffuser was varied to study the lake and sediment oxygen responses to different oxygenation conditions. Here, we calibrated the model to the data for trap height in Singleton et al. (2007), and we validated the model to data for peeling height, trap height, and dissolved oxygen (DO) distribution in Rezvani (2016) and Bierlein et al. (2017).

Model Calibration

Singleton et al. (2007) presented measurements of the dissolved oxygen and temperature profiles in Spring Hollow reservoir on July 2, 2003, August 17, 2003, and October 23, 2004 [Fig. 4(a)]. The reservoir had a diffuser of length 300 m installed at an average depth of 60 m during the 2003 experiments, whereas it had a shorter length of 250 m and depth of 59 m prior to the 2004 experiments [Fig. 3(b)]. The diffuser was constructed from porous hose, yielding a continuous air injection along the length of the diffuser. Flow-rates from the diffuser during field measurements are shown in Table 1. From Spring Hollow reservoir data, we identified the intrusion layer trap height (Figs. 1 and 2) as the zone of excess dissolved oxygen concentration seen in the measured DO profiles; temperature profiles were used to compute the buoyancy frequency.

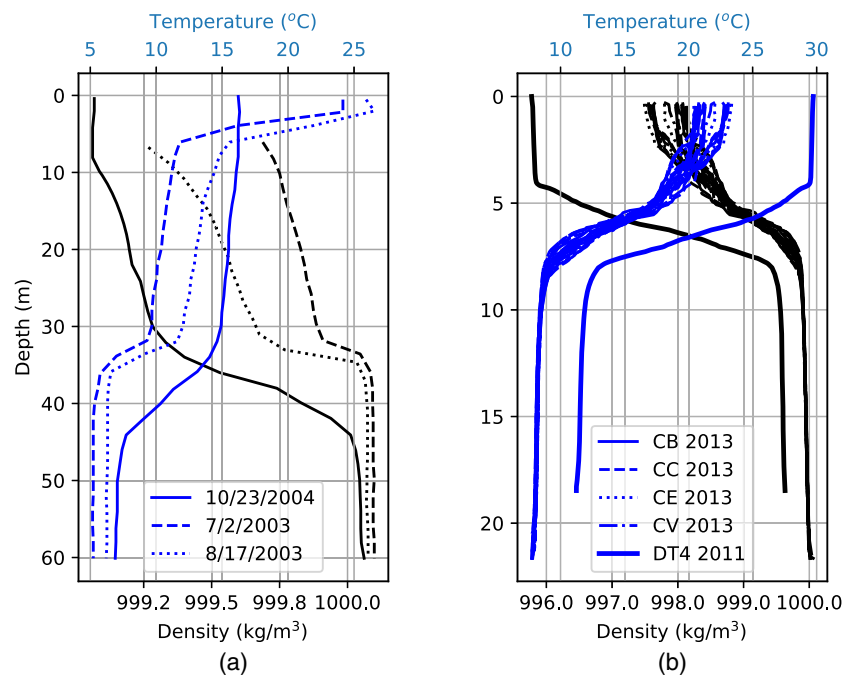


Fig. 4. Measured ambient temperature and density profiles of (a) Spring Hollow Reservoir (data from Singleton et al. 2007); and (b) Carvins Cove Reservoir (data from Rezvani 2016; Bierlein et al. 2017).

Because the lakes were step stratified and N varied with depth, an equivalent buoyancy frequency was calculated, following Lemckert and Imberger (1993) for stepped stratification in lakes to be used in the nondimensionalization process. We simulated these field campaigns and compared the observed and modeled trap heights for different diffuser airflow rates and with different values of ε_p (Fig. 5). To present this model-data comparison in nondimensional space, we took the trap height h_T as the parameter ϕ in Eq. (15).

We compared the model predictions for the neutral buoyant point of the outer plume using four different values of ε_p to the measured field data for trap height (Fig. 5). The model showed weak sensitivity to ε_p due to the step-stratification of these reservoirs. Because the hypolimnion was weakly stratified [Fig. 4(a)], it provided little resistance to the upward-flowing inner plume. As the thermocline was encountered, detrainment was rapid owing to the strong stratification [i.e., $u_i \rightarrow 0$ in Eq. (11)], and reasonable values of ε_p yielded similar results. Because of the weak sensitivity of the model to ε_p , we selected the value $\varepsilon_p = 0.015$, which also matches the calibration value recommended in Socolofsky et al. (2008) for point-source bubble plumes. Using this value, the mean absolute error of the model for nondimensional trap height was 19% (Fig. 5). This level of error is similar to the average 33% error reported for a wide range of plume parameters in Socolofsky et al. (2008) for point-source double-plume integral models. Moreover, mean absolute error of the peel height in calibration cases was 7%.

Model Validation

The field measurement from Rezvani (2016) for Carvins Cove reservoir for experiments in 2011 included measured DO and temperature profiles in the reservoir from July 25 to August 19, 2011, along the intersect ab shown in Fig. 3(b). The reservoir had two bubble plume diffusers installed in parallel, with an average separation distance of 75 m, that ran along a length of 625 m within the deepest (21–23 m), southeastern region of the lake. Like Spring Hollow Reservoir, the diffusers in Carvins Cove were porous

hose with uniform gas discharge along the length of the hose. The diffusers were operated at a total gas flow rate of $15 \times 10^{-6} \text{ Nm}^3\text{s}^{-1} \text{ m}^{-1}$ (Table 1), entering each diffuser and using a gas composition of 98% oxygen and 2% nitrogen. The mean bubble size released from the diffuser was 1 mm (Rezvani 2016). From the measured DO profiles over this period, the plume trap height was about 8.5 m above the bottom. In the model simulations, we modeled one of the two diffusers using a slot width of 15 cm; measurements were generally taken between the two line-plumes of the diffuser system, which were not observed to merge in the field.

For the model validation, we compared the modeled inner and outer plume flow rate data to the measured DO profiles in Carvins Cove reservoir for 2011 (Fig. 6). The diffusers were located near a position 40 m from the bank [Fig. 4(b)], and the intrusion depth was centered between 9 and 12 m [region of DO maximum in Fig. 6(a)]. The model-predicted peeling zone was near a depth of 8 m [Fig. 6(b)], close to the thermocline [Fig. 4(b)], which is the expected behavior in a step-stratified reservoir. The outer plume shrouded over a similar region as the maximum oxygen concentration, and the trap depth predicted by the model,

Table 1. Release flow rates during field experiments in Spring Hollow Reservoir in 2003 and 2004 and Carvins Cove Reservoir in 2011 and 2013

Year of measurement	Date of measurement	Diffuser gas flowrate ($\text{Nm}^3\text{s}^{-1} \text{ m}^{-1} \times 10^{-6}$)
2013	May 27	15.1
	May 28	11.3
	May 29	7.55
	May 30	3.77
	May 31	0
	June 1	5.10
2011	August 1	15.1
2004	October 23	44.4
2003	July 2	35.2
	August 17	12.0

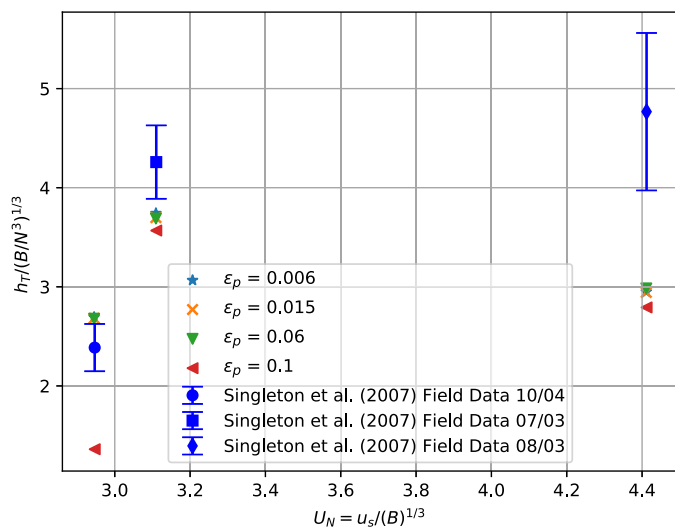


Fig. 5. Comparison of model-predicted trap height for various ε -values (symbols) with observed trap height measured by Singleton et al. (2007).

12.5 m (Table 2), corresponded to the neutral buoyant point of the outer plume. These results demonstrate that the model-predicted bubble plume dynamics corresponded well to the measured features in the reservoir.

Similar measurements in Carvins Cove reservoir were reported in Bierlein et al. (2017) for another measurement campaign in 2013. We used these data to further validate the model performance. The diffusers were operated from May 27 to June 2, 2013 with different gas flow rates each day (Table 1). DO and temperature profiles [Fig. 4(b)] were measured at four locations, namely points CE, CV, CB, and CC [Fig. 3(b)].

For this data set, first, we compared the model-predicted trap height to the measured data at point CV near the middle of the diffusers. During this field campaign, the hypolimnion was fairly well mixed with oxygen so that a weak oxygen maximum between 10

and 16 m depth was apparent over the period of the observations (Figs. 7 and 8). Model predictions for May 27, 2013 predicted a trap depth of 12 m, in reasonable agreement within the maximum dissolved oxygen concentration region (Figs. 7 and 8). The peel height in the model was likewise at 7.5 m depth, close to the thermocline at 8 m depth in the measurements. We also evaluated the model output to determine the mass flux of oxygen into the hypolimnion through dissolution of the gas bubbles and intrusion formation (Fig. 8). About 65% of the released oxygen gas dissolved before reaching the peel height [Fig. 8(b)], and the intrusion was predicted to have a DO concentration of 12 mg L⁻¹ [Fig. 8(a)]. These values also agreed with the field data: a weak gas flux of small bubbles was observed at the surface in the field (i.e., bubbles did not fully dissolve), and the measured oxygen concentration in the DO profiles had a peak between 12 and 13 mg L⁻¹, as seen in the ambient DO profile at the same location [Figs. 7(a) and 8(a)].

Second, we compared the model simulations to the full period of record at the four measurement locations in Carvins Cove reservoir in 2013, and the results are summarized in Table 2. In the simulations, we used the measured temperature profile at each location [Fig. 3(b)] as the ambient stratification. The measured DO maxima lay between the peel-depths and trap-depths predicted by the model for each measurement location, as seen in the case for point CV (Fig. 7). In all the simulations, the modeled concentrations also compared well to the measurements, showing an oxygen-enriched intrusion layer (Table 2). Modeled peel heights were near the thermocline on all days, with a mean absolute error of 9% in peeling height over the validation period. Overall, the predictions for trap height in the validation dataset had a mean absolute error of 24% over the validation period, and we attribute this variation partly to the difficulty in identifying clear trap levels from the measured DO concentration data (Figs. 7 and 8) and to weak stratification below the thermocline that existed in the field but may not have been well captured in the model input data. In weakly stratified conditions, the outer plume tends to shroud the inner plume over a long distance, passing the neutral buoyancy location in the outer plume owing to weak mixing. Hence, the outer plume

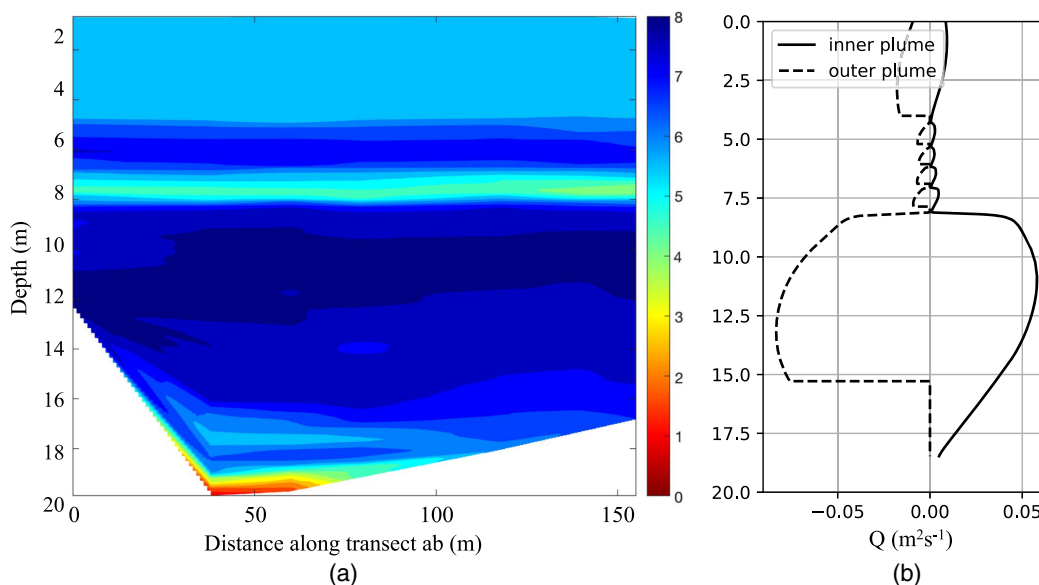
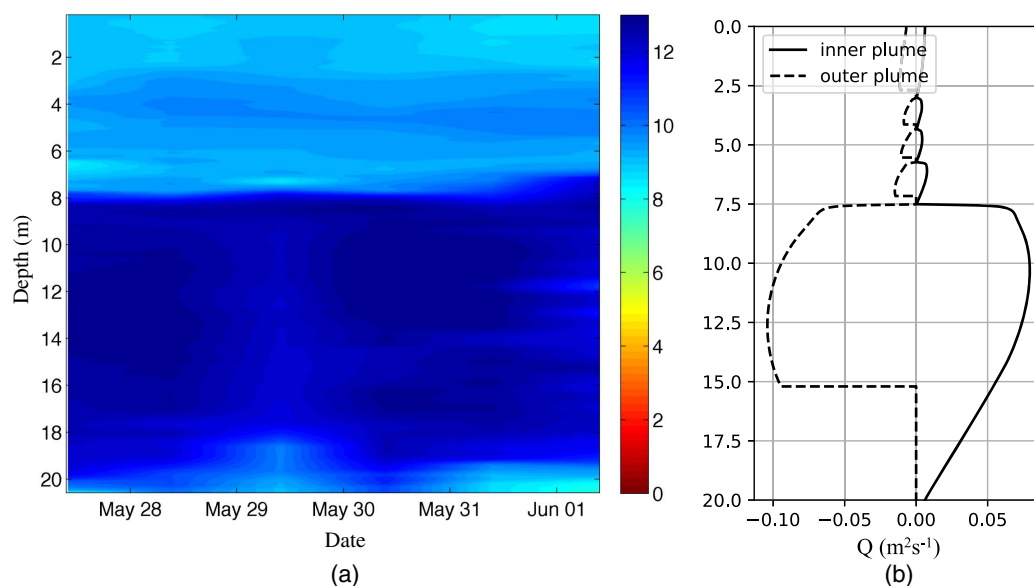


Fig. 6. (a) Contours of measured dissolved oxygen concentration (colorbar, mg L⁻¹) across the diffuser intersect *ab* on August 1, 2011 at Carvins Cove Reservoir; and (b) the variation of flow rate (Q) of the inner and outer plume, predicted from the model using the ambient water column data from August 1, 2011, at location DT4 (79 m).

Table 2. Model-predicted and measured dissolved oxygen concentration at peeling levels and trap levels of the plume at locations CB, CC, CE, CV, and DT4 in Carvins Cove Reservoir

Station	Day	DO concentration (mg L ⁻¹)							
		Peel height (m)		Trap height (m)		At trap height		At peel height	
		Measured	Modeled	Measured	Modeled	Measured	Modeled	Measured	Modeled
DT4	August 1, 2011	10.0 ± 1.0	10.39	8 ± 1.5	5.32	8.0 ± 0.1	7.08	7.8 ± 0.1	7.283
CB	May 27, 2013	11.5 ± 1.0	13.32	8.5 ± 1.5	7.66	12.2 ± 0.1	11.52	11.8 ± 0.1	11.30
	May 28, 2013	11.2 ± 1.0	13.03	8 ± 1.5	8.04	12.1 ± 0.1	11.43	11.7 ± 0.1	11.31
	May 29, 2013	11.5 ± 1.0	13.09	9 ± 2.0	6.13	12.2 ± 0.1	11.23	11.6 ± 0.1	11.64
	May 30, 2013	12 ± 1.0	13.49	10 ± 1.5	5.40	9.0 ± 0.1	10.64	11.8 ± 0.1	12.28
	June 1, 2013	12 ± 1.0	12.46	8 ± 1.5	3.92	9.7 ± 0.1	10.61	11.2 ± 0.1	10.83
CC	May 27, 2013	11 ± 1.0	13.33	8.0 ± 2.0	7.64	7.6 ± 0.1	12.40	11.7 ± 0.1	12.36
	May 28, 2013	11.25 ± 1.0	13.33	8.25 ± 2.5	7.23	8.6 ± 0.1	12.39	11.8 ± 0.1	12.34
	May 29, 2013	11 ± 1.0	12.95	6.1 ± 3.0	6.69	6.5 ± 0.1	11.34	12.4 ± 0.1	10.02
	May 30, 2013	11.75 ± 1.0	13.41	7 ± 3.0	6.49	12.1 ± 0.1	12.47	11.8 ± 0.1	12.95
	June 1, 2013	11.5 ± 1.0	12.74	8 ± 2.0	5.41	12.1 ± 0.1	11.25	11.8 ± 0.1	11.98
CE	May 27, 2013	12.18 ± 1.0	11.92	8 ± 2.0	8.28	12.4 ± 0.1	12.37	12.5 ± 0.1	12.46
	May 28, 2013	12 ± 1.0	11.74	8 ± 2.0	6.61	12.2 ± 0.1	11.80	12.0 ± 0.1	11.74
	May 29, 2013	12.4 ± 1.0	11.54	8.0 ± 2.5	6.68	12.0 ± 0.1	11.57	11.8 ± 0.1	11.50
	May 30, 2013	11.6 ± 1.0	10.47	8.0 ± 2	4.41	12.1 ± 0.1	11.79	11.9 ± 0.1	11.37
	June 1, 2013	11.2 ± 1.0	11.17	8.00 ± 1.5	6.43	11.2 ± 0.1	10.99	10.9 ± 0.1	11.11
CV	May 27, 2013	12.2 ± 1.0	12.67	6 ± 1.5	8.12	12.6 ± 0.1	11.34	12.6 ± 0.1	11.12
	May 28, 2013	12.2 ± 1.0	12.31	6.5 ± 1.5	7.64	12.7 ± 0.1	11.32	12.6 ± 0.1	11.07
	May 29, 2013	12.2 ± 1.0	12.42	7 ± 1.5	5.45	12.0 ± 0.1	12.17	11.6 ± 0.1	12.18
	May 30, 2013	11.4 ± 1.0	12.60	8 ± 1.5	6.59	12.8 ± 0.1	11.96	12.7 ± 0.1	12.34
	June 1, 2013	13.2 ± 1.0	12.14	9 ± 1.5	4.66	11.8 ± 0.1	11.47	11.6 ± 0.1	11.52

**Fig. 7.** (a) Contours of measured dissolved oxygen concentration (colorbar, mg L⁻¹) measured at location CV [Fig. 3(b)], close to the diffusers, from May 27–June 2, 2013 at Carvins Cove Reservoir; and (b) the variation of flow rate (Q) of the inner and outer plume predicted from the model with ambient water column data from May 27, 2013.

intrusion level is more sensitive to the ambient density gradient when it is weak. The DO concentration of the peeling and intruding water levels predicted by the model agreed well with the measurements over the period of the measurement campaign, with a mean absolute error of 5% and 11%, respectively (Table 2). Because these model errors are similar to double-plume integral models for point-source plumes (Socolofsky et al. 2008), we conclude that the present model adequately predicted the behavior of these line-source bubble plumes.

Effect of the Double-Plume Model on Plume Behavior

Integral models of bubble plumes are used in reservoir management for several different purposes. In the case of hypolimnetic oxygenation, the model may be used to design the gas bubble size so that dissolution is largely complete within the hypolimnion and to design the gas flow rate so that minimal reservoir destratification occurs. Dissolution dynamics can largely be evaluated with a

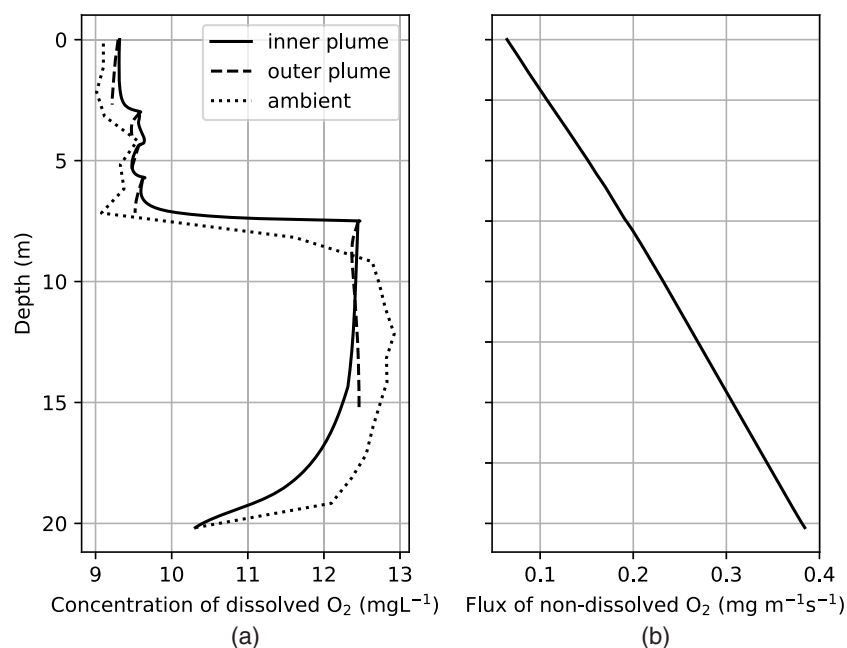


Fig. 8. (a) Predicted dissolved oxygen concentration; and (b) the vertical mass flux of gaseous oxygen in the plume model for May 27, 2013, in Carvins Cove reservoir.

single-plume integral model (i.e., without the outer plume), but to evaluate hypolimnetic mixing, the intrusion formation of the outer plume may be considered. Often, near-field integral plume models are coupled to computational fluid dynamic (CFD) models of the far-field reservoir (though this was not done here). In this case, the entrainment fluxes along the length of the plume and the detrainment flow rate are important sink and source terms for the CFD simulation (e.g., Choi and Lee 2007; Nikiforakis et al. 2017). Previously, single-plume models have been used for this coupling (Singleton et al. 2010; Chen et al. 2017). Choi and Lee (2007) introduced the method “Distributed Entrainment Sink Approach (DESA)” to couple the VISJET (Lee and Chu 2003) Lagrangian plume model with a far-field shallow-water circulation model. In this method, interaction of the plume with the ambient flow is modeled by a distribution of sinks along the plume trajectory and an equivalent source at the plume terminal level, creating a two-way dynamic link between the near and far-field models. Morelissen et al. (2013, 2016) and Horita et al. (2019) presented details of dynamic coupling of a near-field plume with a far-field model for outfall discharges using the DESA method. They coupled the near field plume (CorJet) from the CORMIX model (Jirka 2004, 2006) with Delft3D-FLOW (Lesser et al. 2004). Stamou and Nikiforakis (2013) presented an integrated model coupling the nearfield model CorJet (Jirka 2004, 2006) and a far field model, via an active coupling algorithm that is similar to the DESA method. In their method, the near-field hydrodynamic (velocities) and water quality (temperature) information were transferred to the far field model. Chen et al. (2017) implemented coupling as a series of sinks in a semiimplicit 3D hydrodynamic model to interpret the entrainment of water into a bubble plume model using grid cells within the region between the diffuser depth and the model-estimated maximum plume rise level. Then, at the depth of the maximum plume rise, the discharge was considered as a source to the grid water-jet model coupled with the hydrodynamic model to predict the variation of the thermal structure in a reservoir. In a similar manner, the present model can be coupled with hydrodynamic models with detailed information on the detrainment from the near-field plume model,

which is not included in these previous studies. Here, we compared the predictions of the double-plume integral model to that of a single-plume model to evaluate the differences in the boundary conditions that would be used in model coupling using either of these two approaches.

In this comparison, we considered the data of Singleton et al. (2007) in Spring Hollow reservoir on October 23, 2004 (Fig. 9). When the double-plume model is not considered, the single-plume simulation terminates at the maximum level of plume rise at 42.5 m depth, where its vertical momentum reaches zero and the flow rate of entrained water in the plume is $0.14 \text{ m}^3 \text{ s}^{-1} \text{ m}^{-1}$. If coupled to a CFD model, the entrainment along the total height of the plume would be added as a sink term, and the plume water flux at the top of the plume would be inserted at this maximum height, along with the predicted temperature and oxygen concentration of the single-plume model (Choi and Lee 2007). Within the CFD model, this intruding water from the top of the plume would descend to a neutral level, but without dynamic coupling to the inner plume and likely using a model grid resolution that is very coarse compared to the meter-scale width of the plume. Nonetheless, because both models accurately conserve buoyancy, the intrusion formation may still be acceptable (Zhang and Adams 1999).

When the double-plume model is considered, the peeling height of 42.5 m depth is very close to the maximum height of plume rise of the single plume model (Fig. 9), and the ensuing outer-plume intrusion formation is modeled through dynamic coupling to the inner plume model. This dynamic coupling is achieved by iteration of the inner and outer plume models until their behavior converges. Because the outer plume shrouds the inner plume, the total plume flow rate at the top of the inner plume reduces to $0.1 \text{ m}^3 \text{ s}^{-1} \text{ m}^{-1}$ compared to the single-plume model and flows into the outer plume. By the time the outer plume momentum decays to zero, the total water flow rate in the intrusion increases to $0.15 \text{ m}^3 \text{ s}^{-1} \text{ m}^{-1}$, entering the reservoir at the trap depth of about 50 m. Due to their different dynamics, the double-plume integral model predicts a trap height of 10.0 m, about 40% lower than the height-of-maximum-plume-rise of 16 m predicted by the single-plume model (Fig. 9). Further,

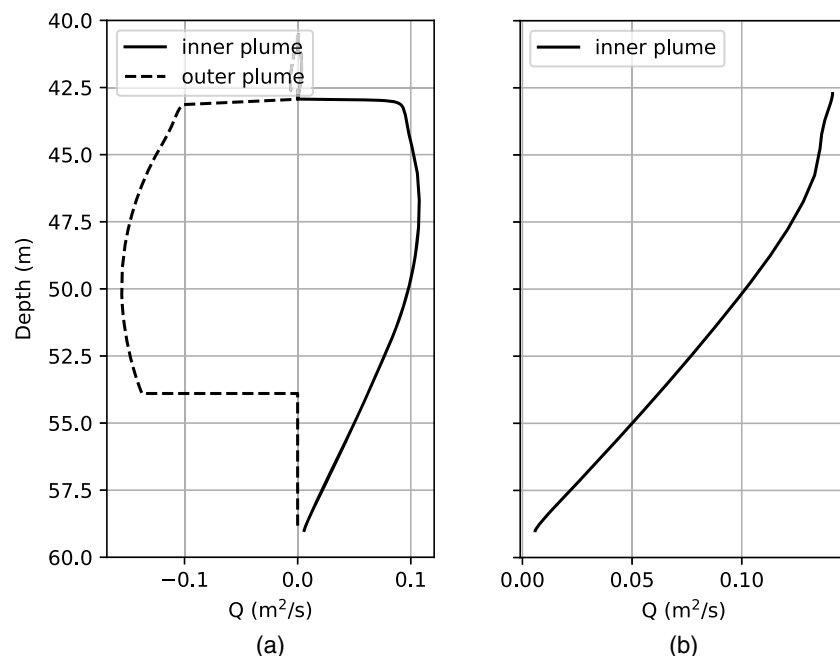


Fig. 9. Plume flow rate (Q) as a function of depth for (a) the double-plume integral model; and (b) the inner-plume model alone in Spring Hollow Reservoir for October 23, 2004, using the ambient water column data from Singleton et al. (2007).

the intrusion flow rate of the double-plume integral model is about double the single-plume model prediction. When coupled to a CFD model, the results from the double-plume integral model are also closer to neutral buoyancy; hence, the double-plume integral model handles the fine-scale, nonhydrostatic mixing in the nearfield of the plume, and the CFD model would simulate the midfield and far-field dynamics without having to adjust to the intrusion formation. Overall, a more stable CFD model solution may be expected.

Summary and Conclusions

In this paper, we developed a top-hat integral model of a bubble plume for line-source geometry by adapting the solution for existing point-source models. We presented the important model equations for the line-source bubble plume using a continuous peeling model for formation of the outer plume. The developed model considered the effects of density stratification and predicted the fate of the gas bubbles, entrained plume water fluxes, intrusion formation, trap height of the intrusion, and oxygen concentration in the intruding water. The resulting model was calibrated to field data for trap height in two reservoirs and validated to field data for trap height and dissolved oxygen concentration in one reservoir. Model performance was similar to that reported for point-source, double-plume integral models of bubble plumes in quiescent stratification. By comparing the double-plume model predictions to the single-plume approach common in the literature for reservoir oxygenation, we showed that by including the outer plume in the near-field model, better dynamic coupling for intrusion formation was achieved and more stable far-field model solutions can be expected than using single-plume models only.

Data Availability Statement

Some or all data, models, or code generated or used during the study are available in a repository or online in accordance with

funder data retention policies. Model simulation code and simulation results will be publicly available through the Gulf of Mexico Research Initiative Information & Data Cooperative (GRIIDC) at <https://data.gulfresearchinitiative.org/10.7266/YTFWPAS8>.

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References

- Asaeda, T., and J. Imberger. 1993. "Structure of bubble plumes in linearly stratified environments." *J. Fluid Mech.* 249: 35–57. <https://doi.org/10.1017/S0022112093001065>.
- Bierlein, K. A., M. Rezvani, S. A. Socolofsky, L. D. Bryant, A. Wüest, and J. C. Little. 2017. "Increased sediment oxygen flux in lakes and reservoirs: The impact of hypolimnetic oxygenation." *Water Resour. Res.* 53 (6): 4876–4890. <https://doi.org/10.1002/2016WR019850>.
- Bombardelli, F., G. Buscaglia, C. Rehmann, L. Rincón, and M. García. 2007. "Modeling and scaling of aeration bubble plumes: A two-phase flow analysis." *J. Hydraul. Res.* 45 (5): 617–630. <https://doi.org/10.1080/00221686.2007.9521798>.
- Cederwall, K., and J. D. Ditmars. 1970. *Analysis of air-bubble plumes*. Pasadena, CA: California Institute of Technology.
- Chen, S., C. Lei, C. C. Carey, P. A. Gantzer, and J. C. Little. 2017. "A coupled three-dimensional hydrodynamic model for predicting hypolimnetic oxygenation and epilimnetic mixing in a shallow eutrophic reservoir." *Water Resour. Res.* 53 (1): 470–484. <https://doi.org/10.1002/2016WR019279>.
- Choi, K., and J. H. Lee. 2007. "Distributed entrainment sink approach for modeling mixing and transport in the intermediate field." *J. Hydraul. Eng.* 133 (7): 804–815. [https://doi.org/10.1061/\(ASCE\)0733-9429\(2007\)133:7\(804\)](https://doi.org/10.1061/(ASCE)0733-9429(2007)133:7(804)).

- Crounse, B., E. Wannamaker, and E. Adams. 2007. "Integral model of a multiphase plume in quiescent stratification." *J. Hydraul. Eng.* 133 (1): 70–76. [https://doi.org/10.1061/\(ASCE\)0733-9429\(2007\)133:1\(70\)](https://doi.org/10.1061/(ASCE)0733-9429(2007)133:1(70)).
- Dissanayake, A. L., J. Gros, and S. A. Socolofsky. 2018. "Integral models for bubble, droplet, and multiphase plume dynamics in stratification and crossflow." *Environ. Fluid Mech.* 18 (5): 1167–1202. <https://doi.org/10.1007/s10652-018-9591-y>.
- Dissanayake, A. L., M. Rezvani, S. A. Socolofsky, K. A. Bierlein, and J. C. Little. 2016. "Integral model for bubble plumes from line-source geometry." In *Proc., Int. Symp. on Outfall Systems*. New York: Springer.
- Fanneløp, T., S. Hirschberg, and J. Küffer. 1991. "Surface current and recirculating cells generated by bubble curtains and jets." *J. Fluid Mech.* 229: 629–657. <https://doi.org/10.1017/S0022112091003208>.
- Fanneløp, T., and K. Sjoen. 1980. *Hydrodynamics of underwater blowouts: Ship research*. Trondheim, Norway: Institute of Norway, Marine Technology Centre.
- Fanneløp, T., and D. Webber. 2003. "On buoyant plumes rising from area sources in a calm environment." *J. Fluid Mech.* 497: 319–334. <https://doi.org/10.1017/S0022112003006669>.
- Fox, D. G. 1970. "Forced plume in a stratified fluid." *J. Geophys. Res.* 75 (33): 6818–6835. <https://doi.org/10.1029/JC075i033p06818>.
- Gill, A. E. 1982. Vol. 30 of *Atmosphere-ocean dynamics*. New York: Academic Press.
- Harleman, D. R. F., G. H. Jirka, and D. H. Evans. 1973. *Experimental investigation of submerged multiport diffusers for condenser water discharges with application to the Northport electric generating station*. Technical Rep. No. 164. Boston: Massachusetts Institute of Technology.
- Horita, C. O., T. B. Bleninger, R. Morelissen, and J. L. B. de Carvalho. 2019. "Dynamic coupling of a near with a far field model for outfall discharges." *J. Appl. Water Eng. Res.* 7 (4): 295–313. <https://doi.org/10.1080/23249676.2019.1685413>.
- Hussain, N., and B. Narang. 1984. "Simplified analysis of air-bubble plumes in moderately stratified environments." *J. Heat Transfer* 106 (3): 543–551. <https://doi.org/10.1115/1.3246713>.
- Jirka, G. H. 2004. "Integral model for turbulent buoyant jets in unbounded stratified flows. Part I: Single round jet." *Environ. Fluid Mech.* 4 (1): 1–56. <https://doi.org/10.1023/A:1025583110842>.
- Jirka, G. H. 2006. "Integral model for turbulent buoyant jets in unbounded stratified flows. Part II: Plane jet dynamics resulting from multiport diffuser jets." *Environ. Fluid Mech.* 6 (1): 43–100. <https://doi.org/10.1007/s10652-005-4656-0>.
- Kaminski, E., S. Tait, and G. Carazzo. 2005. "Turbulent entrainment in jets with arbitrary buoyancy." *J. Fluid Mech.* 526: 361–376. <https://doi.org/10.1017/S0022112004003209>.
- Kobus, H. E. 1968. "Analysis of the flow induced by air-bubble systems." In *Proc., 11th Int. Conf. Coastal Engineering*, 1016–1031. Reston, VA: ASCE.
- Lee, J. H., and V. H. Chu. 2003. Vol. 1 of *Turbulent jets and plumes: A Lagrangian approach*. New York: Springer.
- Lee, S. L., and H. W. Emmons. 1961. "A study of natural convection above a line fire." *J. Fluid Mech.* 11 (3): 353–368. <https://doi.org/10.1017/S0022112061000573>.
- Lemckert, C. J., and J. Imberger. 1993. "Energetic bubble plumes in arbitrary stratification." *J. Hydraul. Eng.* 119 (6): 680–703. [https://doi.org/10.1061/\(ASCE\)0733-9429\(1993\)119:6\(680\)](https://doi.org/10.1061/(ASCE)0733-9429(1993)119:6(680)).
- Lesser, G. R., J. V. Roelvink, J. A. T. M. Van Kester, and G. S. Stelling. 2004. "Development and validation of a three-dimensional morphological model." *Coastal Eng.* 51 (8–9): 883–915. <https://doi.org/10.1016/j.coastaleng.2004.07.014>.
- Liedtke, J., and M. Schatzmann. 1997. *Dispersion from strongly buoyant sources*. Final Rep. EU-Project EV5V-CT-93-0262. Hamburg, Germany: Univ. of Hamburg, Meteorological Institute.
- McDougall, T. J. 1978. "Bubble plumes in stratified environments." *J. Fluid Mech.* 85 (4): 655–672. <https://doi.org/10.1017/S0022112078000841>.
- McGinnis, D., J. Little, and A. Wüest. 2001. "Hypolimnetic oxygenation: Coupling bubble-plume and reservoir models." In *Proc., Asian WATERQUAL*. Fukuoka, Japan: International Water Association.
- McGinnis, D., A. Lorke, A. Wüest, A. Stöckli, and J. Little. 2004. "Interaction between a bubble plume and the near field in a stratified lake." *Water Resour. Res.* 40 (10): 1–11. <https://doi.org/10.1029/2004WR003038>.
- Milgram, J. H. 1983. "Mean flow in round bubble plumes." *J. Fluid Mech.* 133: 345–376. <https://doi.org/10.1017/S0022112083001950>.
- Morelissen, R., T. van der Kaaij, and T. Bleninger. 2013. "Dynamic coupling of near field and far field models for simulating effluent discharges." *Water Sci. Technol.* 67 (10): 2210–2220. <https://doi.org/10.2166/wst.2013.081>.
- Morelissen, R., R. Vlijm, I. Hwang, R. L. Doneker, and A. S. Ramachandran. 2016. "Hydrodynamic modelling of large-scale cooling water outfalls with a dynamically coupled near-field-far-field modelling system." *J. Appl. Water Eng. Res.* 4 (2): 138–151. <https://doi.org/10.1080/23249676.2015.1099480>.
- Morton, B. R., G. I. Taylor, and J. S. Turner. 1956. "Turbulent gravitational convection from maintained and instantaneous sources." *Proc. R. Soc. London, Ser. A*. 234 (1196): 1–23.
- Nikiforakis, I. K., A. I. Stamou, and G. C. Christodoulou. 2017. "Integrated modeling of single port brine discharges into unstratified stagnant ambient." *Environ. Fluid Mech.* 17 (1): 247–275. <https://doi.org/10.1007/s10652-016-9473-0>.
- Priestley, C. H. B., and F. K. Ball. 1955. "Continuous convection from an isolated source of heat." *Q. J. R. Meteorol. Soc.* 81 (348): 144–157. <https://doi.org/10.1002/qj.49708134803>.
- Rezvani, M. 2016. "Bubble plumes in crossflows: Laboratory and field measurements of their fluid dynamic properties with application to lake aeration and management." Ph.D. dissertation, Zachry Dept. of Civil Engineering, Texas A&M Univ.
- Ricou, F. P., and D. Spalding. 1961. "Measurements of entrainment by axisymmetrical turbulent jets." *J. Fluid Mech.* 11 (1): 21–32. <https://doi.org/10.1017/S0022112061000834>.
- Rouse, H., C. S. Yih, and H. Humphreys. 1952. "Gravitational convection from a boundary source." *Tellus* 4 (3): 201–210. <https://doi.org/10.3402/tellusa.v4i3.8688>.
- Schladow, S. 1992. "Bubble plume dynamics in a stratified medium and the implications for water quality amelioration in lakes." *Water Resour. Res.* 28 (2): 313–321. <https://doi.org/10.1029/91WR02499>.
- Singleton, V. L., P. Gantzer, and J. C. Little. 2007. "Linear bubble plume model for hypolimnetic oxygenation: Full-scale validation and sensitivity analysis." *Water Resour. Res.* 43 (2): 1–12. <https://doi.org/10.1029/2005WR004836>.
- Singleton, V. L., F. J. Rueda, and J. C. Little. 2010. "A coupled bubble plume-reservoir model for hypolimnetic oxygenation." *Water Resour. Res.* 46 (12): W12538. <https://doi.org/10.1029/2009WR009012>.
- Socolofsky, S. A., and E. E. Adams. 2005. "Role of slip velocity in the behavior of stratified multiphase plumes." *J. Hydraul. Eng.* 131 (4): 273–282. [https://doi.org/10.1061/\(ASCE\)0733-9429\(2005\)131:4\(273\)](https://doi.org/10.1061/(ASCE)0733-9429(2005)131:4(273)).
- Socolofsky, S. A., T. Bhaumik, and D.-G. Seol. 2008. "Double-plume integral models for near-field mixing in multiphase plumes." *J. Hydraul. Eng.* 134 (6): 772–783. [https://doi.org/10.1061/\(ASCE\)0733-9429\(2008\)134:6\(772\)](https://doi.org/10.1061/(ASCE)0733-9429(2008)134:6(772)).
- Stamou, A. I., and I. K. Nikiforakis. 2013. "Integrated modelling of single port, steady-state thermal discharges in unstratified coastal waters." *Environ. Fluid Mech.* 13 (4): 309–336. <https://doi.org/10.1007/s10652-012-9266-z>.
- van Reeuwijk, M., and J. Craske. 2015. "Energy-consistent entrainment relations for jets and plumes." *J. Fluid Mech.* 782: 333–355. <https://doi.org/10.1017/jfm.2015.534>.
- Wüest, A., N. H. Brooks, and D. M. Imboden. 1992. "Bubble plume modeling for lake restoration." *Water Resour. Res.* 28 (12): 3235–3250. <https://doi.org/10.1029/92WR01681>.
- Yang, D., B. Chen, S. A. Socolofsky, M. Chamecki, and C. Meneveau. 2016. "Large-eddy simulation and parameterization of buoyant plume dynamics in stratified flow." *J. Fluid Mech.* 794: 798–833. <https://doi.org/10.1017/jfm.2016.191>.
- Yuan, L.-M., and G. Cox. 1995. "Entrainment of line thermal plumes." *Fire Saf. Sci.* 2 (1): 200–209.
- Zhang, X. Y., and E. E. Adams. 1999. "Prediction of near field plume characteristics using far field circulation model." *J. Hydraul. Eng.* 125 (3): 233–241. [https://doi.org/10.1061/\(ASCE\)0733-9429\(1999\)125:3\(233\)](https://doi.org/10.1061/(ASCE)0733-9429(1999)125:3(233)).